

Acknowledgments

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I am deeply grateful to Kenny Woo Shi Nam, a Year 5 student at the HKBU School of Chinese Medicine (cGPA 3.97/4.0), currently interning at Guangdong Provincial Hospital of TCM. Kenny’s contributions were extensive: he advised on the selection of the 17 classical texts, conducted rigorous testing across formula retrieval, clause comparison, and clinical case analysis, and provided detailed feedback on diagnostic accuracy and system usability that directly shaped multiple system improvements.

I also thank the doctoral students at Guangzhou University of Chinese Medicine, several of whom also practice as physicians at Guangdong Provincial Hospital of TCM, who participated in the Arena evaluation and provided clinical feedback, including specific recommendations for corpus expansion and insights on bridging classical theory with modern clinical practice.

AI-powered development environments (OpenCode, Cursor) were used for code implementation assistance throughout the project, with all code reviewed and tested by the author. AI writing tools were used for grammar and style refinement during report preparation. All architectural decisions, system design, and written content are the author’s own work.

Abstract

Traditional Chinese Medicine (TCM) practitioners face significant challenges in retrieving and synthesizing evidence from a vast corpus of classical literature. Existing Large Language Models (LLMs) often function as “black boxes,” providing claims that lack traceable links to authoritative source material. This project introduces TCM-Sage, an evidence-synthesis tool designed to provide transparent, citation-backed insights from 17 foundational TCM texts comprising 3.72 million characters and 12,204 indexed chunks. By implementing a Retrieval-Augmented Generation (RAG) pipeline with hybrid retrieval—combining DashScope text-embedding-v4 vector search with the SymMap v2.0 knowledge graph (18,450 entities)—TCM-Sage ensures every claim is traceable to its original source. Evaluation via a blind A/B Arena system with 59 votes from TCM students and practitioners yielded a paired t-test result of $t = 3.44$, $p = 0.0011$ ($p < 0.01$), and Cohen’s $d = 0.45$ (medium effect), demonstrating statistically significant preference for the RAG system over a general-purpose LLM with web search.

Chapter 1

Introduction

1.1 Background and Motivation

Traditionally, practitioners and educators of Traditional Chinese Medicine (TCM) mainly rely on the classical literature for both evidence-based clinical practice and theoretical foundation. The TCM classical literature is an extensive historical document spanning two millennia of medical practice and development. While there are 17 fundamental texts totalling over 3.7 million characters of classical Chinese language texts, the task of evidence-retrieval and cross-text analysis for these complex historical documents is very time-consuming and challenging, even for the most experienced professionals and educators in the field. In order to enhance TCM education and clinical practice, it is essential to have computational support.

This project was inspired by my dad's suggestion to look into the intersection of AI and medicine, and my friends who are studying with a focus in Traditional Chinese Medicine. I had the chance to bounce my ideas off of Kenny Woo Shi Nam, a Year 5 student at the HKBU School of Chinese Medicine with a cGPA of 3.97/4.0, who is currently interning at the Guangdong Provincial Hospital of TCM. He emphasized the need for a tool that can extract and provide citations from these classical texts, because currently available digital resources are not deep enough, and existing tools are too siloed to be sufficient for in-depth studies of traditional Chinese medicine.

I actually wanted to do a system that recommends courses to students based on their interest and learning style but it later evolved into something much more complex and significant that's building an AI tool for Traditional Chinese Medicine. The proposal got approved by my supervisor Dr. Zhang Ce and it's a pretty demanding project.

TCM-Sage is based on the assumption that AI can function as external memory for a human practitioner. While a human cannot possibly memorize all the wordy clauses throughout the 17 texts analyzed for this project, a Retrieval-Augmented Generation (RAG) system can recall 100

Table 1.1: The 17 Classical TCM Texts Included in TCM-Sage

Category	Included Texts
Foundational Theory	Huangdi Neijing Suwen, Nanjing
Shanghan/Wenbing	Shanghan Lun, Jingui Yaolue, Wenbing Tiaobian, Wenre Lun
Internal Medicine	Beiji Qianjin Yaofang, Neiwaishang Bianhuo Lun
Acupuncture	Huangdi Neijing Lingshu, Zhenjiu Jiayijing
Herbology/Formulary	Shennong Bencao Jing, Bencao Gangmu
Jin-Yuan Four Masters	Suwen Xuanji Yuanbing Shi, Rumen Shiqin, Piwei Lun, Lanshi Micang, Danxi Xinfu

1.2 Problem Statement

Although there is an imperative need to integrate AI into the practice and future development of Traditional Chinese Medicine (TCM), there remains the challenge of verifying claims made by AI programs with reliable sources of information. Modern AI programs are sometimes able to provide citations for the information that they generate, and some can even search the internet for citations when requested. However, the user cannot verify the accuracy of the information retrieved from the internet, and thus cannot verify the citations that the

AI program provides. Unlike AI programs, TCM clinicians require reliable citations to specific classical sources in the TCM classics and other authoritative sources. Getting a prescription or treatment wrong, whether due to incorrect dosage or an incorrect formula, can have serious consequences in clinical practice.

A RAG system can get close to zero hallucination if all of its output is grounded in source content that was actually retrieved. In the Arena blind testing, the general-purpose LLM made up information and stated it with full confidence — much of it entirely fabricated. An early version of TCM-Sage also got confused at the clause level, returning *Shanghan Lun* Clause 2 content when asked about Clause 82. That turned out to be a chunking granularity problem rather than a fundamental flaw. After clause-level retrieval was added, TCM-Sage pulled up the right clause every time, whereas the general-purpose LLM kept hallucinating with no way to self-correct.

TCM AI products like HuatuoGPT and iFlytek’s medical AI work as “AI Doctors”, which can directly make diagnoses and provide prescriptions. Unlike TCM-Sage which uses an evidence synthesis methodology supported by traceable citations from ancient TCM texts, the “AI Doctor” approach faces several structural problems.

1. Data Sovereignty - concerns around the upload of patient data to third-party cloud-based systems from hospitals and practitioners.
2. **Clinical Risk:** High liability involved in providing a direct diagnosis; lack of nuanced TCM *Bian Zheng Lun Zhi* (Pattern Differentiation and Treatment) thinking.
3. There is no Evidence Synthesis for *Classical Sources with Clause-Level Citations*.

It feels like there is missing functionality in the current suite of evidence synthesis tools: support for the diagnostic work performed by a human expert, without letting the AI do the expert’s work for them. Ideally such “Glass Box” functionality would display all claims made by the AI, along with sources cited (if any).

1.3 Proposed Solution

Unlike many of the new Evidence-Synthesis Tools that are appearing, TCM-Sage is not an “AI Doctor” – rather it is a “Glass Box” which enables the user to view all the data that is used in the synthesis. All the information is traceable back to the specific page in one of the 17 classical texts where the information can be reviewed. TCM-Sage is there to support TCM practitioners by providing synthesis of evidence that can be used to inform their decision making for a case or topic that the practitioner is researching.

The solution uses Retrieval-Augmented Generation (RAG) with Hybrid Retrieval.

- **Vector Search:** We used DashScope’s text-embedding-v4 model to search for the most similar chunks of text within our database.
- Integration with Knowledge Graph: Leverage the data contained in the SymMap v2.0 knowledge graph to bridge the semantic gap between current medical names and classical Chinese medical descriptors.

While TCM-Sage can provide treatment suggestions when full “Si Zhen” (Four Examinations) case data is provided, all recommendations are grounded in the literature and are intended for educational purposes only. This design principle ensures that the system remains a reference tool, maintaining the practitioner’s role in the diagnostic process while providing the necessary evidence to support their conclusions.

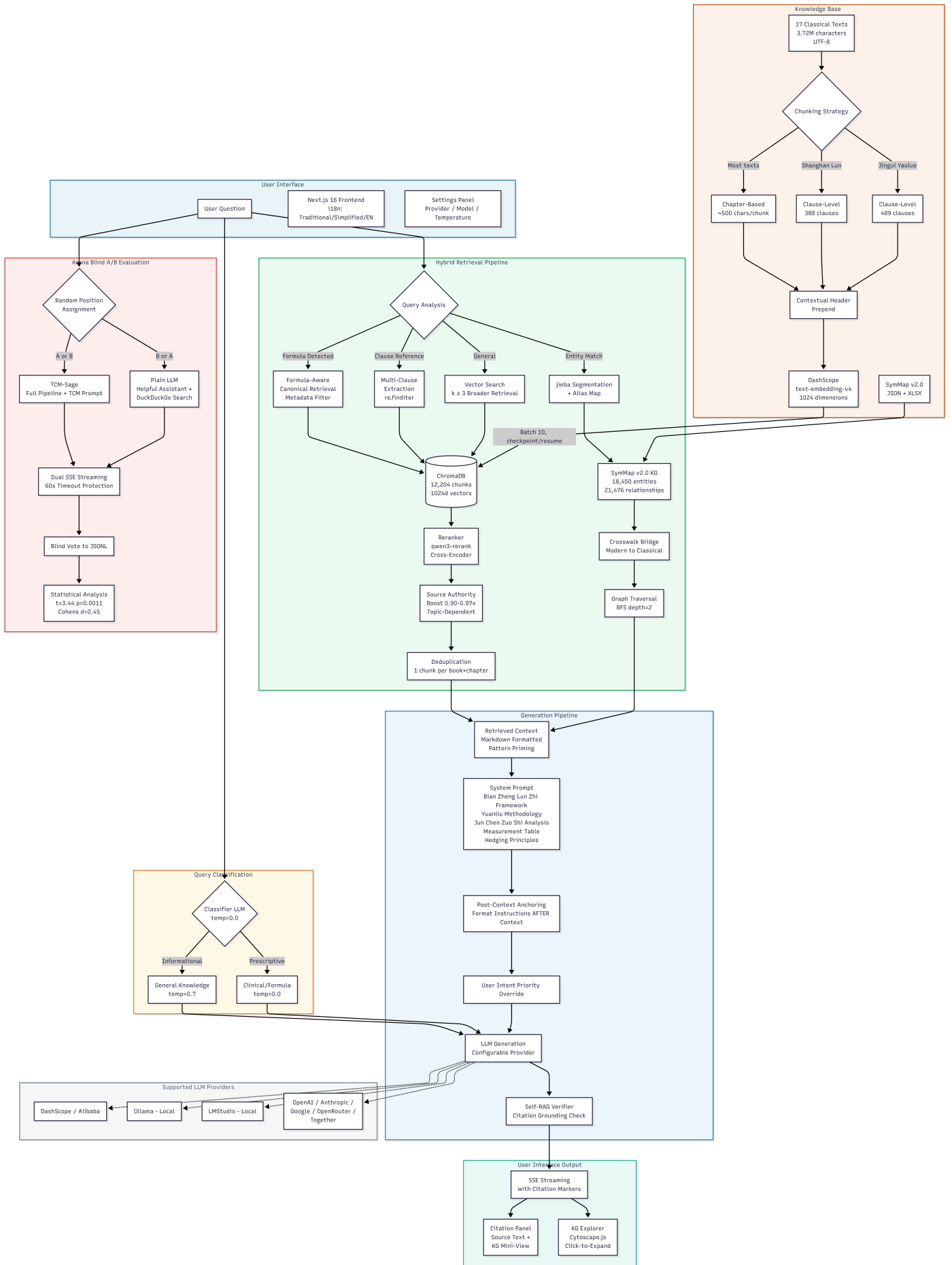


Figure 1.1: High-level architecture of the TCM-Sage system.

1.4 Objectives

The project aims to achieve the following technical and functional objectives:

1. Design and implement a relatively modular pipeline for handling classical Chinese medical texts, and fine-tune this pipeline for the 17 individual text sets, bearing in mind their particular linguistic characteristics.
2. **Knowledge Graph Integration:** Load the massive SymMap v2.0 knowledge graph which contains 18,450+ concept entities and 21,476 connections between them. Then, create a crosswalk (for advanced use cases) to link modern symptoms to historical or classical entities for enhanced analysis capabilities.
3. **Clause-Level Chunking of Key Texts:** Chunking of analyzed key texts. The Shanghan Lun is chunked into 388 clause-chunks and the Jingui Yaolue is chunked into 489 clause-chunks. Thus, citations can point to the appropriate section of a key text.
4. **Arena Evaluation System.** Develop a blind A/B evaluation system modelled after the LMSYS Chatbot Arena web site, through which the statistical validity of the RAG output can be verified using the standard paired t-tests and Cohen’s d effect sizes.
5. **Local Deployment Support:** Support fully local deployment (on laptop or desktop) of both LLM inference with Ollama or LMStudio and of embedding models in DashScope. Everything in DashScope defaults to using cloud APIs, but every component is substitutable with an offline version that can be used in a completely offline workflow.
6. Have a very visible way to view and separately navigate citations, such as a tab in the frontend interface. This would allow users to double click on individual claims to be taken to the relevant source and read from wherever that claim was extracted from (e.g. a particular chapter or section). It would also show information about the object being cited.

1.5 Scope and Limitations

The core knowledge base consists of 17 classical TCM texts on TCM theory, acupuncture, Shanghan/Wenbing and herbology, as well as the Jin-Yuan Four Masters. This forms the main corpus of data that has been encoded into the knowledge base. In addition to this, the LLM will have learned general knowledge not contained in the data and tagged by the built-in verifier with : “not supported by retrieved sources” .

The current corpus does not include books and articles on areas such as surgery, orthopedics, ophthalmology, specialized gynecology and pediatrics monographs and pulse diagnosis books. These areas can be included in future development of the corpus. This system is designed for referencing evidence and should NOT be used for clinical decision making.

Given the output is in language form, the upper limit of its quality would be around the quality of the LLM it was generated with. Even though the RAG pipeline structures the questions with proper context, it still depends on the model to properly understand the input in classical Chinese, and to generate decent human-sounding modern Chinese or English output. That being said, there is still some capacity to fine-tune even small LLMs, such as Qwen 2.5 Turbo with the RAG pipeline to generate fairly decent outputs.

1.6 Report Structure

The remainder of this report is organized as follows:

- This chapter provides an overview of the state of the art of RAG technology, TCM AI methods and related applications, and literature of knowledge graph in medical informatics. (Chapter 2 Literature Review)
- Chapter 3: System Architecture and Design. This chapter details the RAG pipeline as well as the enhanced data structure and functionality provided by SymMap v2.0. It also discusses the crosswalk bridge implementation.
- Chapter 4: Implementation - A behind the scenes look at the FastAPI backend, the Next.js frontend and my local deployment configuration.

- **Chapter 5: Evaluation.** Results from the Arena blind A/B testing experiment and results of the statistical analysis of these results for the evaluated system.
- **Chapter 6: Discussion and Contributions** - Evaluation of the strengths and weaknesses of the system; Discussion of the project's contributions; Comments from the practitioners involved and a discussion of the technical problems that were encountered.
- **Chapter 7: Conclusion and Future Work.** This chapter summarizes the results obtained in this work and outlines future work directions to increase the size of the needed corpus and to improve the retrieval effectiveness.

Chapter 2

Literature Review

This chapter provides a theoretical framework discussion and reviews the current state-of-the-art research done towards the development of TCM-Sage. It discusses on the current state-of-the-art of Retrieval-Augmented Generation models, TCM related AI systems and applications, Knowledge Graph in medical informatics and evaluation methodologies for language model outputs.

2.1 Retrieval-Augmented Generation (RAG)

Due to the disadvantages caused by static Large Language Models, such as the lack of updating knowledge after “knowledge cutoff” and the problem of hallucinations, where models generate entirely false but realistic content, models based on Retrieval-Augmented Generation (RAG) have been increasingly used. A RAG model consists of a pre-trained generator and a retrieval module [Lewis et al. \(2020\)](#).

2.1.1 RAG Fundamentals and Architecture

RAG works in two stages. In the retrieval stage, the system pulls relevant documents from a non-parametric knowledge base that matches the user’s query. These retrieved documents are then attached to the front of the user’s prompt before it goes to the generator, giving it a grounded context for producing its output. What makes this powerful is that up-to-date or niche knowledge can be plugged in without having to retrain or fine-tune the model.

2.1.2 RAG vs. Fine-Tuning Trade-offs

While fine-tuning a pre-trained model to adapt it to new tasks and languages in a specific domain is known as weight adaptation, RAG offers additional benefits particularly relevant for medical applications. Using weight adaptation methods, such as LoRA and QLoRA, significant reduction in catastrophic forgetting was observed during fine-tuning. Although freezing the weights of the pre-trained language models enables the network to generalize to new instances and selectively invoke different paths of processing within the model via active adapter, fine-tuning on limited amounts of in-domain data can still cause a reduction in generalization performance. Weight adaptation is intended to change a model’s *behavior* (e.g., output format, response style), not to reliably inject new *knowledge*—attempts to do so often introduce hallucination or overwrite existing capabilities.

TCM-Sage adopts a RAG-first approach for four primary reasons. First, fine-tuning on classical Chinese texts is prohibitively expensive and may yield poor results due to the scarcity of high-quality instruction-tuning pairs. Second, the RAG knowledge base can be updated or expanded by simply adding new text chunks to the vector store, requiring no retraining. Third, the modular nature of RAG allows the base LLM to be swapped as newer, more capable models become available. Finally, this architecture provides a foundation for future hybrid approaches where a fine-tuned TCM model can be paired with a RAG pipeline for maximum precision.

2.1.3 Advanced RAG Techniques

There has been growing interest in “beyond naive RAG”. Reranking plays important roles in boosting the retrieval quality and precision. A reranker model evaluates the initial set of retrieved documents and re-orders them based on semantic relevance, ensuring that the most pertinent information appears at the top of the context window [Gao et al. \(2024\)](#).

Self-RAG and verification techniques further enhance reliability by allowing the model to critique its own retrieved context and generated output [Asai et al. \(2024\)](#). In domain-specific applications like medicine, these verification loops are essential for ensuring that the synthesized advice aligns with the source material. MedRAG research highlights the unique challenges of medical retrieval, including the need for high-precision terminology matching and the handling of complex clinical reasoning [Xiong et al. \(2024\)](#).

2.2 TCM Knowledge Representation and Existing Systems

Knowledge representation and Artificial Intelligence (AI) applications in Traditional Chinese Medicine (TCM) are more challenging than ordinary medical applications. The valuable knowledge stored in the enormous number of classical medical books spanning over two millennia forms the backbone of TCM. While there are rich resources of such knowledge, the chief obstacle for their computerized capture lies in the very different way they are written in classical Chinese as opposed to contemporary Chinese used in practicing medicine today.

2.2.1 Classical Chinese Text Challenges

Classical Chinese is characterized by extreme brevity and high semantic density. A single character often carries multiple meanings depending on the context (one-character-multiple-meanings). Archaic grammar and semantic drift across different dynasties further complicate automated analysis. For example, a term used in the *Huangdi Neijing* (approx. 300 BCE) may have a different clinical implication in the *Wenbing* literature of the Qing Dynasty (1644-1911 CE).

2.2.2 Existing TCM AI Products

Several research institutions and corporations have developed TCM-specific AI systems [Wang et al. \(2026\)](#). HuatuoGPT, developed by CUHK Shenzhen, achieved state-of-the-art results on multiple medical benchmarks [Zhang et al. \(2023\)](#). Qihuang 2.0 (Shuzhi Qihuang), a 32-billion-parameter multimodal model developed by East China Normal University in collaboration with multiple medical and research institutions, scored 88.1% on the TCM licensing examination and incorporated tongue and pulse image analysis [East China Normal University School of Pharmacy \(2024\)](#). iFlytek’s TCM diagnostic assistant was deployed across 107 primary care institutions in Bozhou, Anhui Province, serving over 350 TCM practitioners [China National Radio \(CNR\) \(2024\)](#). These systems share a common design philosophy: direct diagnosis and prescription.

2.2.3 The “AI Doctor” Gap

Existing TCM AI systems have primarily pursued the “AI Doctor” paradigm, focusing on direct diagnosis and prescription [Wang et al. \(2024\)](#). TCM-Sage identifies a different gap: no existing tool focuses on evidence-synthesis from classical texts with traceable, clause-level citations. By prioritizing the “Glass Box” approach, TCM-Sage serves as a reference tool that supports, rather than replaces, the human expert.

2.3 Knowledge Graphs in Medical AI

Knowledge Graphs (KGs) provide a structured, symbolic representation of domain knowledge that complements the sub-symbolic nature of LLMs. In medical AI, KGs ground the model’s output in established facts and relationships.

2.3.1 SymMap v2.0 Integration

SymMap v2.0 is a comprehensive TCM knowledge graph containing 18,450 entities and 21,476 relationships [Wu et al. \(2019\)](#). It integrates modern pharmacology with traditional herbology, providing a bridge between different medical paradigms. TCM-Sage uses SymMap to enhance retrieval by identifying related symptoms, herbs, and formulas that may not be explicitly mentioned in the user’s query but are semantically linked in the TCM theoretical framework.

2.3.2 Bridging Modern and Classical Terminology

A major challenge in TCM AI is the semantic drift between modern medical terminology and classical descriptions. A practitioner might use a modern term for a symptom that appears in the *Shanghan Lun* under a different name. While entity alignment (synonym resolution within TCM terminology) and biomedical entity linking (mapping clinical text to ontologies like SNOMED) are established techniques, the specific task of bridging a modern structured knowledge graph to a classical text corpus remains largely unexplored [Xiang et al. \(2025\)](#). TCM-Sage addresses this gap through a crosswalk bridge that maps SymMap’s modern entities to classical text chunks, enabling bidirectional retrieval between modern queries and classical evidence.

2.4 Evaluation Methodologies

Evaluating the quality of LLM-generated medical advice is difficult due to the subjective and nuanced nature of clinical reasoning. Traditional metrics like ROUGE or BLEU are insufficient for assessing factual accuracy or clinical relevance.

2.4.1 LMSYS Chatbot Arena and Blind A/B Testing

The LMSYS Chatbot Arena has popularized blind A/B testing as a widely recognized benchmark for evaluating LLM performance [LMSYS Org \(2024\)](#). In this framework, users compare two anonymized outputs and vote for the superior response. This method captures human preferences and nuances that automated rubrics miss.

2.4.2 Statistical Validation

To ensure that the results of blind A/B testing are scientifically valid, TCM-Sage employs standard statistical methods. The paired t-test determines whether the difference in scores between the RAG system and a baseline LLM is statistically significant. Cohen’s d quantifies the effect size, indicating the practical magnitude of any improvement. Together, these methods provide a statistical foundation for evaluating system performance beyond anecdotal evidence.

Chapter 3

System Architecture and Design

This chapter describes the design of the technical architecture for TCM-Sage and discusses some of the principal design criteria. The subsequent chapters then progress from the present system, through the design and construction of a special knowledge base, to the design of a hybrid retrieval method and generation pipeline tailored to process classical Chinese medical texts.

3.1 System Overview

The TCM-Sage system is modular and based on a Retrieval-Augmented Generation, evidence-based approach to process classical Traditional Chinese Medicine (TCM) literature. The high-level workflow of the TCM-Sage system consists of a classical text processing step that results in a populated knowledge base, hybrid vector and graph retrieval, and finally a Large Language Model generates new text that synthesizes the results of the prior steps with a specially formatted citation panel for verification purposes.

3.1.1 Three-Phase Evolution

The full three-phase developmental history of TCM-Sage is presented from the inception of the project.

- **Phase 1: Naive RAG.** Our first prototype was evaluated on a single base text, *Huangdi Neijing*. We used a light-weight all-MiniLM-L6-v2 embedding with dimension 384, and performed basic vector search. Surprisingly, we were able to find a few hundred chunks of classical Chinese texts that could be leveraged for LLM synthesis.
- **Phase 2: Hybrid RAG.** In transforming the core architecture in Section 1, we had planned for Phase 2 to include Hybrid RAG that utilizes token embeddings as well as structure information. We utilized the SymMap 2.0 knowledge graph and a novel Self-RAG verifier that not only checks for semantic similarity but also connects modern symptoms to classic entities.
- **Phase 3: Production Hybrid RAG.** The final phase scaled the system to 17 classical texts and upgraded the technical stack. It implemented DashScope text-embedding-v4 (1024 dimensions) and the qwen3-rerank model for high-precision retrieval. This phase also introduced the crosswalk bridge for entity resolution and the full Arena evaluation system.

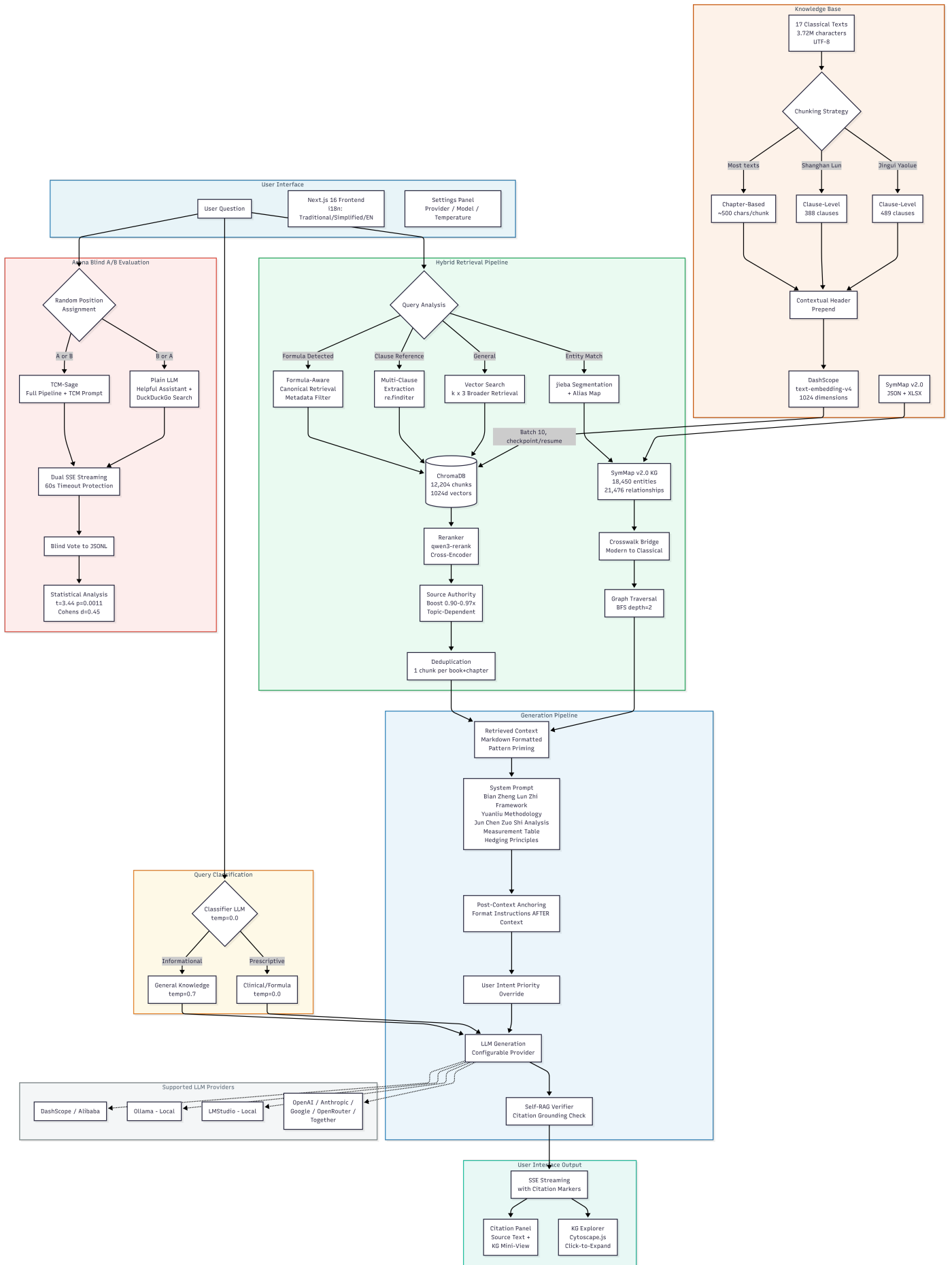


Figure 3.1: High-level architecture of the TCM-Sage production system.

3.2 Knowledge Base Design

The knowledge base of TCM-Sage contains 17 classical TCM books with a total of 3.72 million characters. The knowledge base is organized into 12,204 chunks by applying domain specific strategies for chunking.

3.2.1 Chunking Strategies

A critical feature of the TCM-Sage resource is that the information has been organized into bite-sized ‘chunks’ that correspond to the structure of individual books. Most medical books are arranged in a hierarchical order such as chapters, sections and paragraphs. For the two canonical clinical texts, the *Shanghan Lun* and *Jingui Yaolue*, however, the TCM-Sage has used a yet finer approach for the organization of the information on the clause level.

3.2.2 Clause-Level Chunking

The classic texts *Shanghan Lun* include 388 clauses and *Jingui Yaolue* consist of 489 clauses, each acting as natural semantic units and supplying diagnosis and prescription in the form of brief clinical rules that can be precisely quoted.

To improve embedding quality, formula-aware contextual headers are prepended to each clause. For example, Clause 35 of the *Shanghan Lun* is stored with the header “Shanghan Lun Di 35 Tiao Mahuang Tang:.. This technique ensures that the embedding model captures both the clause number and the primary formula, significantly improving retrieval accuracy for specific clinical queries.

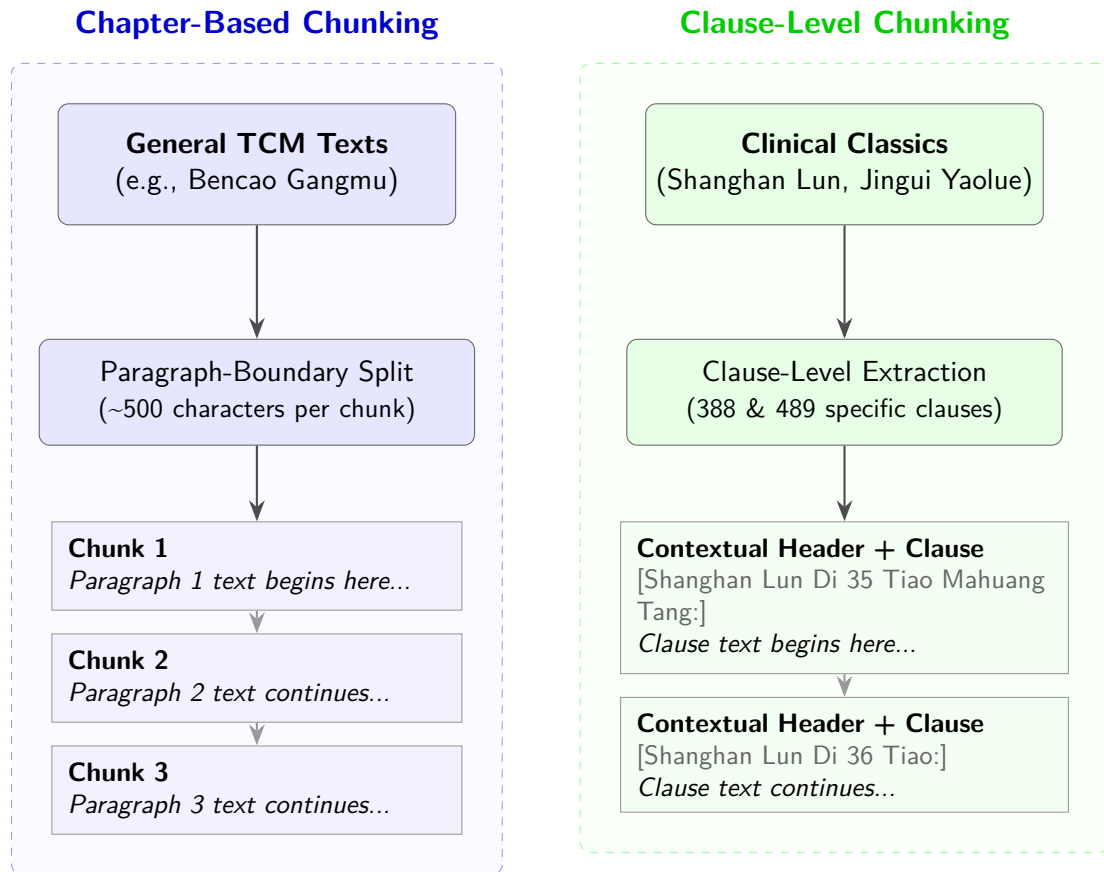


Figure 3.2: Comparison of chunking strategies for different text types.

3.3 Hybrid Retrieval Architecture

We present a novel retrieval framework which couples efficient dense vector search with fast symbolic knowledge graph traversal in order to achieve high recall and high precision.

3.3.1 Vector Search and Chronological Boost

We employ the text-embedding model DashScope text-embedding-v4 (dimension 1024) to embed the semantics of a query and documents into vectors and then perform vector search to get a set of preliminary results. We first retrieve a somewhat expanded set of $k \times 3$ documents, and then apply the cross-encoder DashScope qwen3-rerank to re-rank this set in order to improve semantic ranking before applying further filters.

In contrast to the rest of the paper, for this specific collection (TCM-Sage) the source authority is taken into account when calculating the distance scores. These scores have then been post-processed to generate smooth and consistent scores across topics. The scaling factor for scores of individual queries ranges from 0.90 to 0.97, and these scores are ranked per topic and not linearly. This means that for theoretical queries like "organs" the earliest TCM classic, the Huangdi Neijing, would receive the highest lift, whereas for febrile disease queries the more recent Wenbing works would receive the highest lift.

3.3.2 Formula-Aware Canonical Retrieval

If TCM-Sage spots a specific formula name in the user's query, it kicks off a metadata-filtered search targeting canonical texts. The goal is to make sure the primary source for that formula is always present in the retrieved context, even when other semantically related but less authoritative texts also come up in the results.

3.3.3 Knowledge Graph and Crosswalk Bridge

We then traverse the graph created by SymMap 2.0 using NetworkX. We apply entity matching to process the human language of the query terms: We use the jieba segmentation tool to split the words in the query term into individual characters or word pieces, and also incorporate a custom alias map that contains many modern Chinese synonyms that are mapped to their classical language equivalents. This allows the processing of modern symptom names, which are then mapped to their classical names in the knowledge graph, and herbs, formulas and syndromes that would not otherwise be retrieved by vector search.

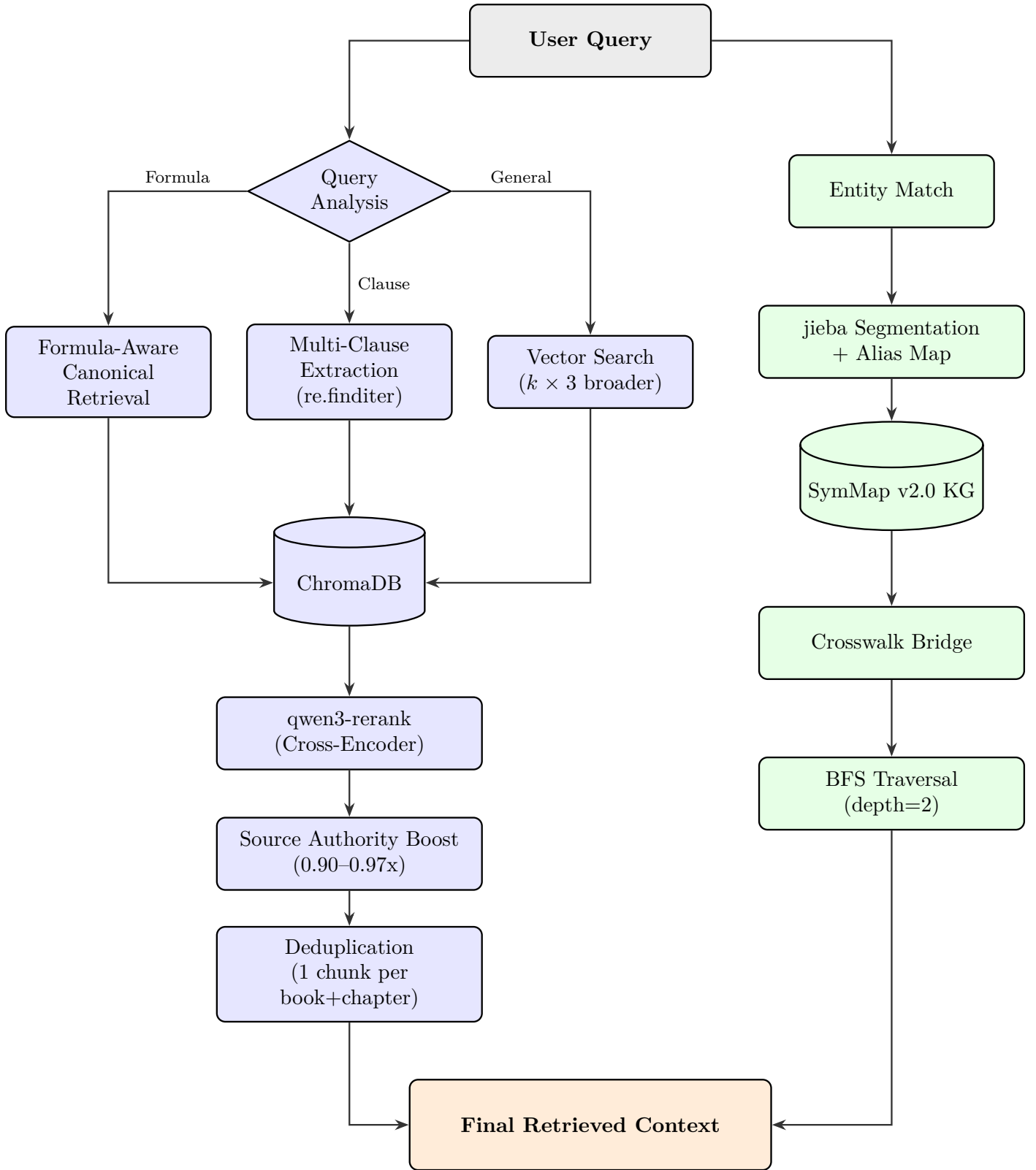


Figure 3.3: The hybrid retrieval pipeline combining vector and graph-based methods.

3.4 Generation Pipeline

We then run the generation pipeline on this context to generate evidence-backed output. All context retrieval providers are supported in this step (DashScope/Alibaba, OpenAI, Anthropic, Google, OpenRouter, Together, Ollama, and LMStudio), and all can be deployed on local machines or in the cloud. We provide a unified factory function, `create_llm()`.

3.4.1 Query Classification and Temperature Control

Before generation, the initial classifier LLM is run at temperature 0.0 for a deterministic outcome. The classifier differentiates between two types of queries, “informational” and “prescriptive”, where informational queries aim to retrieve general knowledge, definitions and explanations and prescriptive queries are directed towards obtaining diagnoses, treatments and formula recommendations. This classification influences the generation temperature:

- **Informational:** Temperature 0.7 — allows natural language generation for informational content.
- **Prescriptive:** Temperature 0.0 — for clinical queries, most accurate and reproducible when highest precision is needed.

This dual-temperature design reflects a safety-conscious approach: clinical queries demand the highest possible fidelity to source material, while general knowledge queries benefit from slightly more natural language generation.

3.4.2 System Prompt and Framework

The system prompt follows the “Bian Zheng Lun Zhi” (Pattern Differentiation and Treatment) and “Yuan Liu” (Source and Development) logic. In addition, for specific “Jun Chen Zuo Shi” (Sovereign, Minister, Assistant, Courier) formulas, pertinent drug information will be provided in the system prompt.

3.4.3 Post-Context Instruction Anchoring

To prevent the model from ignoring formatting directives, TCM-Sage employs post-context instruction anchoring. Instead of placing formatting rules before the retrieved context (where the model tends to forget them), we put them after. That way the last thing the model sees before generating is the output structure we want it to follow.

3.4.4 Pattern Priming and User Intent

Our system primes the LLM to use the retrieved context by outputting the result in a formatted form that encourages the LLM to use headers and blockquotes. We also included an ancient measurement conversion table in the prompt to ensure the proper “dosage”. The “User Intent Priority” line was also added so that if the user has preferences on how the LLM will override the default formatting of the document, those will be prioritized.

3.5 Arena Evaluation Design

Building on the success of the LMSYS Chatbot Arena, we have developed a new system which enables blind A/B testing of TCM Practitioners within our existing evaluation system.

3.5.1 Blind A/B Testing

We compare the full TCM-Sage pipeline (the RAG system) against a plain LLM baseline. For the baseline, we used the same underlying model but gave it a generic “helpful assistant” prompt. It has access to web search (DuckDuckGo). Both responses are streamed out using Server-Sent Events (SSE), and we randomly assign them to the left or right side of the screen so the voter cannot tell which is which.

3.5.2 Statistical Validation

Results of arena votes were tested for significance using a paired t-test and corroborated with Cohen’s d effect size. These results provide proof of concept for the RAG system, showing that observed improvements are not by chance.

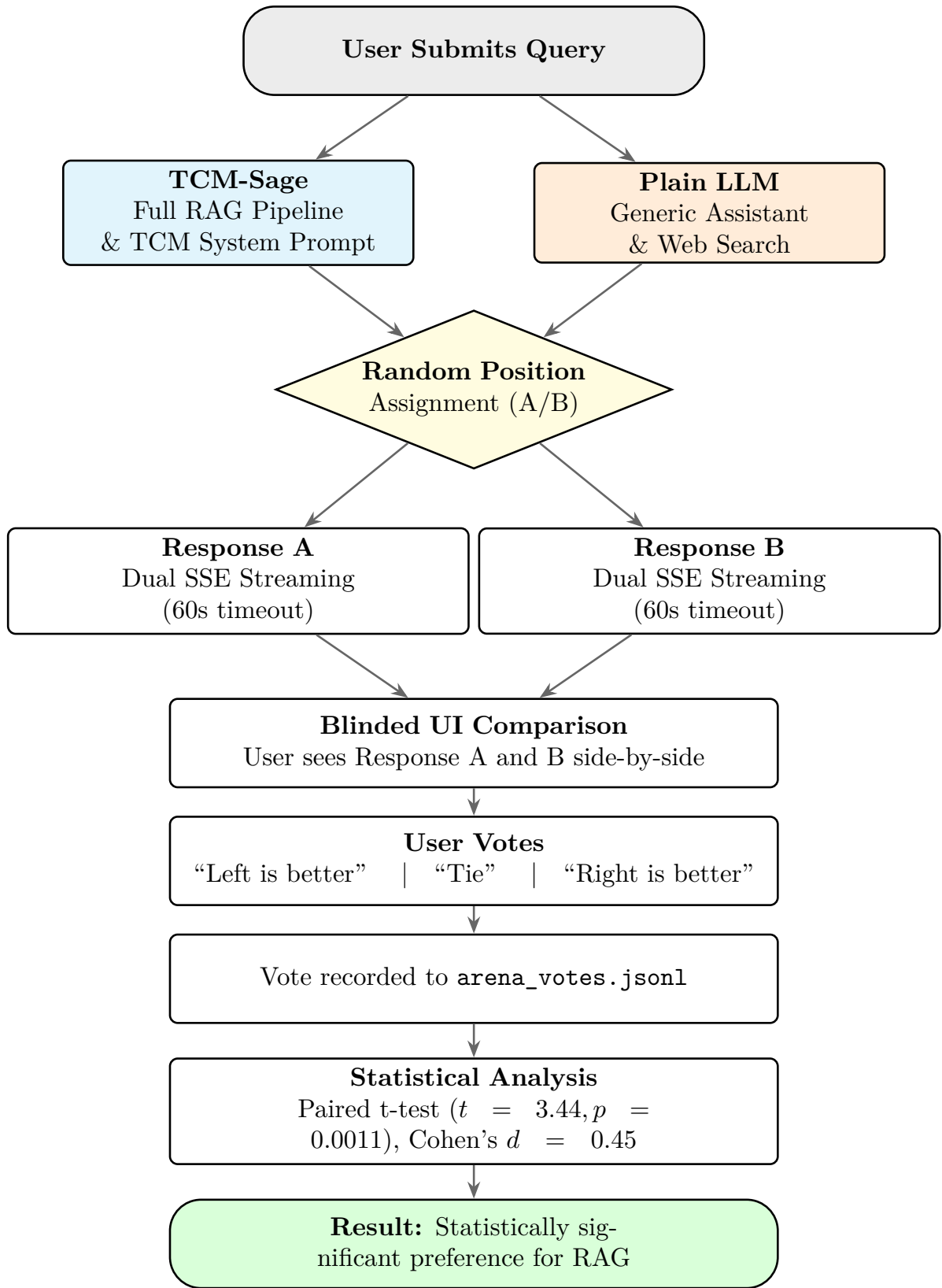


Figure 3.4: The blind A/B Arena evaluation and statistical validation flow.

Chapter 4

Technical Implementation

This chapter outlines the technical details behind the implementation of TCM-Sage. The implementation follows a three-legged approach, namely a retrieval-augmented generation (RAG) pipeline fine-tuned on classical TCM literature, knowledge graph (KG) entity structures, and a framework for blind A/B evaluations called Arena.

4.1 Technology Stack

The system is built on a full-stack architecture with the following components:

4.1.1 Backend Infrastructure

This interface is built with Python 3.13 and utilizes FastAPI to build high-concurrency asynchronous API endpoints. The backend orchestration is handled with the LangChain library which greatly simplifies the process of building retrieval chains as well as crafting prompt templates. We currently utilize ChromaDB, an open-source vector database with support for local persistence, to store vectorized text. This also allows for efficient filtering based off of metadata.

4.1.2 Frontend and User Interface

The user interface is developed with **Next.js 16** (App Router) and **React 19**. **TypeScript** ensures type safety across the application, while **Tailwind CSS** is used for responsive, utility-first styling. The interface supports a dual-stream architecture for the Arena evaluation system, allowing simultaneous rendering of two independent Large Language Model (LLM) responses.

4.1.3 Language Models and Embeddings

- The system utilizes the **DashScope API** (Qwen series) **LLM** function, connected to the international endpoint `dashscope-intl.aliyuncs.com`.
- **Embedding Model:** DashScope text-embedding-v4 (dimensionality: 1024) with TCM domain-specific asymmetric prefixes. The prefix 'Generate semantic representation for this classical TCM text for retrieval' was used for model ingestion, and 'Generate semantic representation for this TCM clinical question to retrieve relevant classical passages' for query embedding. The asymmetric embedding method follows best practices as outlined in DashScope for retrieval purposes dealing with specific domains.
- **Reranker:** `qwen3-rerank` cross-encoder integrated in the `hybrid_search()` pipeline. It is applied *before* combining vector search results but after vector search scores have been computed because the semantic reranking improves the quality of ranking for entity-based queries. Both scores are saved in the metadata for later usage where it is necessary. KG facts (structured entity relationships) do not benefit from semantic reranking.

4.1.4 Knowledge Graph and Local Support

Our entity resolution framework is built on top of the SymMap v2.0 graph generated with NetworkX. We built a custom crosswalk bridge to resolve colloquial / modern medical terms to classical TCM entities. For

researchers requiring local privacy, the system supports local LLM deployment via **Ollama** and **LMStudio** through standard OpenAI-compatible endpoints.

4.2 Data Acquisition and Preprocessing

A RAG system is only as good as the corpus on which it has been trained. For TCM-Sage we have used a collection of 17 classical texts on Traditional Chinese Medicine. The total size of the corpus is 3.72 million characters of UTF-8 encoded text.

4.2.1 Chunking Strategies

For the purpose of semantic accuracy two different methods of chunking have been used for the classical texts.

1. **Chapter-based Chunking:** Most of the corpus is then processed using Chapter-based Chunking. This approach divides the large amount of text in the corpus using the ubiquitous Pianming (chapter title) marker to produce smaller segments of roughly 500 characters in length that are well suited for generating theoretical, open-ended questions.
2. **Clause-level Chunking:** We have especially addressed this task for two classic TCM medical texts, the *Shanghan Lun*, containing 388 clauses, and the *Jingui Yaolue* with 489 clauses. We have employed a regex pattern matching technique on the numerical markers (“N.”) separating the individual clinical clauses for formula-based retrieval. We have defined symptom-sign clusters within a single clause.

4.2.2 Contextual Header Prepend

To further improve both the quality of the embedding and the quality of the retrieval, we introduce a novel, processing-stage sensitive, Contextual Header for each processed chunk of natural language. In this work, we utilize the headers to extract source book and formula names and embed these into the vector space representation of each chunk of natural language. Here, the excerpt “<Shanghan Lun> Clause 35 Mahuang Tang: [Clause Content]” is a short clause of the classic Chinese medical text *Shanghan Lun*, referencing a variation on the Mahuang Tang formula.

4.2.3 Ingestion and Storage

Our ingestion pipeline handles the DashScope limit of 10 embeddings per API call by checkpointing and resuming the ingestion of the API call as necessary. The resulting corpus was split up into chunks of varying size () and amounted to 12,204 total chunks or 850MB of ChromaDB storage.

4.3 Retrieval Pipeline Implementation

Our retrieval pipeline takes into account the classical TCM literature with its notion of authority and formula specific contents.

4.3.1 Formula-Aware Canonical Retrieval

Standard vector search often misses the primary source text of a formula. We added a detection layer to address this:

1. Process formula names (such as *Guizhi Tang*) found in user input.
2. Perform a metadata filtered search of ChromaDB using a metadata-filtered query restricted to canonical texts (*Shanghan Lun* and *Jingui Yaolue*).
3. These canonical results should come before any general results.

4.3.2 Multi-Clause Comparison

When working as a TCM practitioner, it is often necessary to compare different clauses in order to differentiate syndromes. TCM practitioners use clauses to compare and differentiate syndromes. Within this system there is a function called `extract_clause_references`. This function utilises the `re.finditer` method to search for clauses. It is able to identify citations that include citations cited with or without Section/Clause and also non-adjacent clauses. It is also able to extract out multiple clauses with a single call.

4.3.3 Source Authority Boost

A source chronological boost, `SOURCE_CHRONOLOGICAL_BOOST`, is a dictionary-based weight that applies multipliers (ranging from 0.90 to 0.97) on the distance portion of the score of older authoritative sources such as texts from the Han Dynasty rather than simply granting them a blanket boost. As a result, very old sources will score higher than much later, wordier sources of the Ming and Qing Dynasties.

4.3.4 Deduplication and Filtering

...to maximize the diversity of information in the context window.

- **Broad retrieval:** We retrieve $k \times 3$ results before applying truncation post-retrieval.
- **Source Deduplication:** Most relevant excerpt per *book + chapter* unit is included in the corpus to prevent overloading the LLM with very similar excerpts from the same source.
- **ChromaDB Filters:** Complex filtering uses the `$and` operator with nested conditions rather than a flat dictionary; otherwise the ChromaDB filter parser ignores the constraint. The flat dictionaries are ignored by the ChromaDB filter parser.

4.4 Knowledge Graph Integration

We present a knowledge graph for medical information retrieval, which we have compiled into a structured data resource, SymMap v2.0, to supplement the system's ability to search the literature using unstructured text.

4.4.1 Graph Architecture

There are 18,450 entities and 21,476 relationships in the constructed knowledge graph. As shown in figure above, besides Herbs, the three most frequent types of entities are Symptoms, TCM Syndromes (Zheng) and Modern Diseases.

4.4.2 Entity Matching and Crosswalk

Mapping modern user queries to classical graph entities is done through:

- **Jieba Segmentation:** Perform tokenization on the query and extract all the candidate medical terms.
- **Alias Mapping:** Crosswalk bridges the modern Chinese language terms to TCM terms. *du zi teng* (belly ache) $\rightarrow$ *fu tong*, *shui bu zhao* (can't sleep) $\rightarrow$ *shi mian*.
- **BFS Traversal:** For each matched entity, the script performs a BFS traversal with an adjustable depth (default is 2) to extract relevant information such as symptoms or herbs for the LLM to generate diagnostic responses with.

4.5 Generation and Prompt Engineering

The generation layer is guided to encourage the LLM to behave in a professional and natural TCM way, while simultaneously enforcing strict formatting of the output. 8 different LLM cloud providers are supported in the code including a simple factory function `create_llm()` that allows for easy switching between different cloud APIs (DashScope, OpenAI, Anthropic, Google, OpenRouter, Together) and local inference providers (Ollama, LMStudio).

4.5.1 Query Classification

The incoming query is then classified by the classifier LLM (temperature 0.0) to either informational or prescriptive using the function `get_query_severity()`. Then the classified query is passed to the pre-trained LLM corresponding to the classification:

- `llm_informational`: temperature 0.7 for general knowledge queries
- `llm_prescriptive`: temperature 0.0 when dealing with clinical or formula queries

The classifier, main LLMs, and verifier can all be pointed at separate providers and models each, with providers and models settable via environment variables. This means different cost/quality tradeoffs can be made independently for each component.

4.5.2 System Prompt Architecture

I wrap `ChatPromptTemplate.from_messages()` around system and user messages, rather than trying to pack them into a single string to cut down on the highly penalized parameter counts of overly complex string formatting.

The `DEFAULT_SYSTEM_PROMPT` incorporates several professional frameworks:

- Bian Zheng Lun Zhi– diagnosis and treatment differentiation.
- Yuanliu Methodology: tracing the historical roots of formulas found in the different sources and studying their development.
- Jun Chen Zuo Shi: Analysis of drug roles within formulas (Sovereign, Minister, Assistant, Courier).
- Measurement Conversion: We hard code a table for historical weights during the Han Dynasty to correct for a LLM error regarding *liang*. For example, 1 *liang* $\approx$ 13.8g, not 3g as the LLM states.

4.5.3 Context Anchoring and Pattern Priming

One approach to the general problem of context drift is to literally type out a boundary marker like “=== End of References ===”, and then include the instructions in the “[Formatting Requirements]” section after the context has been retrieved. It appears the LLM anchors to the instructions last seen by it, rather than the context. See notes on “Pattern Priming” for this. Also, interestingly the LLM automatically mirrors the clean Markdown output by using headers and blockquotes without being explicitly told to do so.

4.6 Arena Evaluation System

In order to evaluate the RAG approach, Blind A/B testing was performed using a framework implemented in a file called `arena.py`.

4.6.1 Baseline Configuration

We also evaluate a “Plain LLM” baseline that uses the helpful assistant prompt and has access to DuckDuckGo Search (`ddgs` package) like prior work, but passes both the assistant prompt and the search query to the model at once. This is a representative general-purpose configuration for non-specialized models.

4.6.2 Asynchronous Dual Streaming

We use `asyncio`. `Queue` and `asyncio.gather` to generate concurrently both a TCM-Sage response and a baseline response and return them simultaneously. Random position assignment (A/B) is used to counter voter attempt to determine treatment in the two simultaneously displayed responses.

4.6.3 Reliability and Storage

To avoid locking the UI when a backend goes unresponsive, we use 60-second timeouts per stream. All interaction data (query, responses, and final vote) is saved to a stream of JSONL with corresponding timestamps to allow for further statistical analysis.

4.7 Web Application Implementation

The frontend was developed as a professional clinical software with a focus on clarity and traceability.

4.7.1 Internationalization and State

Traditional Chinese (default), Simplified Chinese and English languages are supported. The language state is managed via React Context API and the translations are read from static JSON files.

4.7.2 UX Enhancements

- Smart Auto-Scroll: Our chat will automatically scroll down for you as messages come in, but you can always scroll up and unlock that feature. We'll even show a little “down” button to take you back to the latest messages.
- Citation Panel: By clicking on a citation number, one can view cited sources in a fully readable Citation Panel on the side of the window. Additionally, each cited source is also displayed in a mini knowledge graph format similar to that of an entity click, powered by xyflow/React Flow.
- KG Explorer: There is a full-page graph explorer at `/kg/[entityId]`, where you can browse the knowledge graph, expand nodes by clicking on them, or search for a particular entity. The graph is rendered using the wonderful **Cytoscape.js** library.



Figure 4.1: Citation Panel showing the original source text with click-through to full passage.

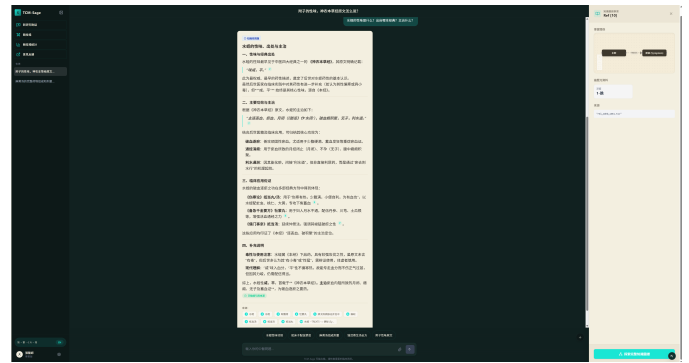


Figure 4.2: Citation Panel showing the mini knowledge graph view for an entity.

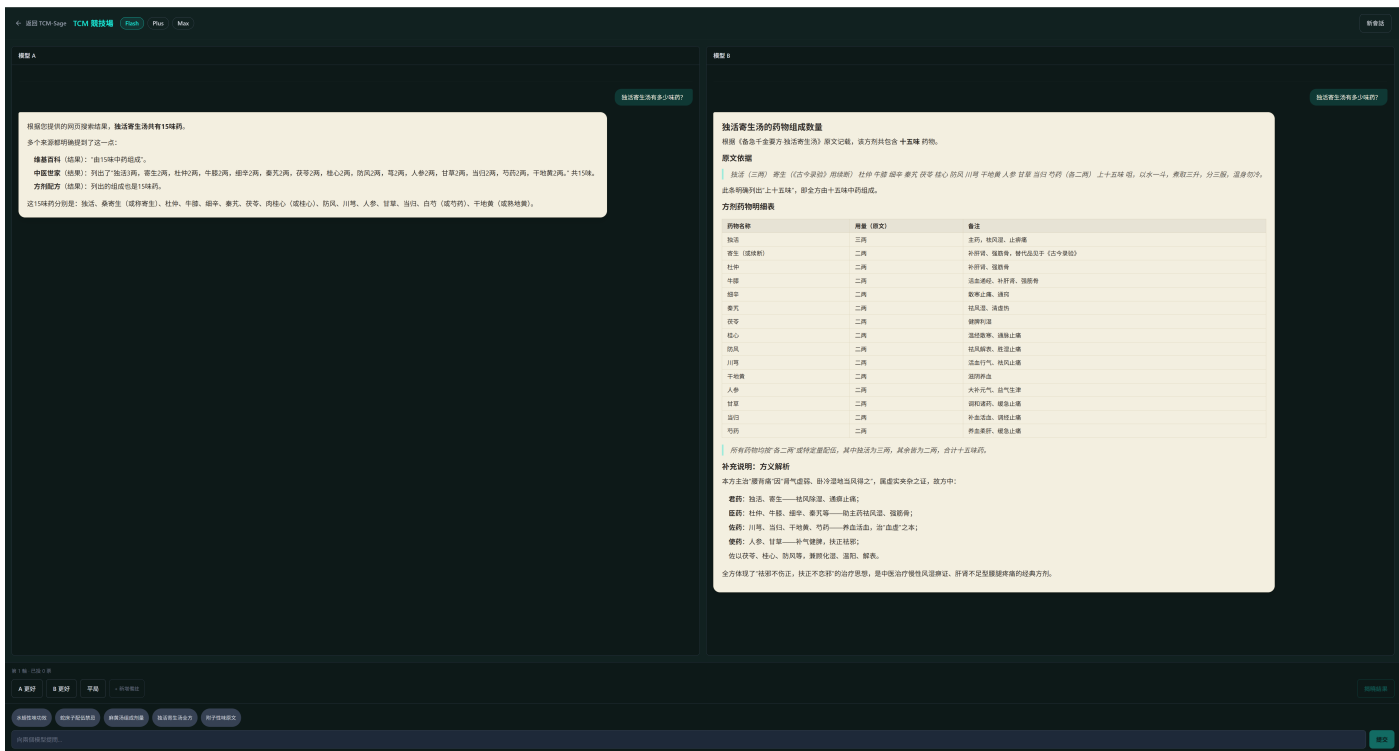


Figure 4.3: The Arena blind A/B evaluation interface with two simultaneous response streams.

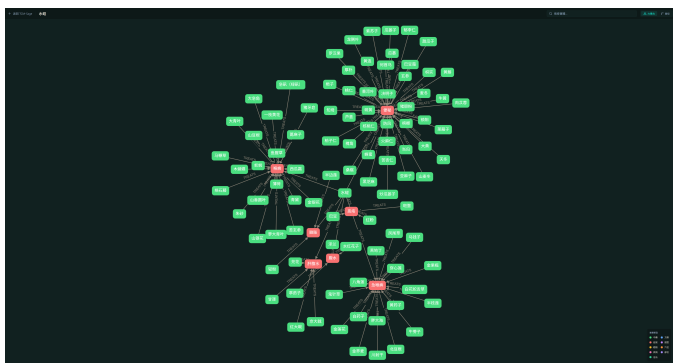


Figure 4.4: Full-page Knowledge Graph explorer with entity search and click-to-expand.



Figure 4.5: Full source text viewer showing the original classical passage with metadata.

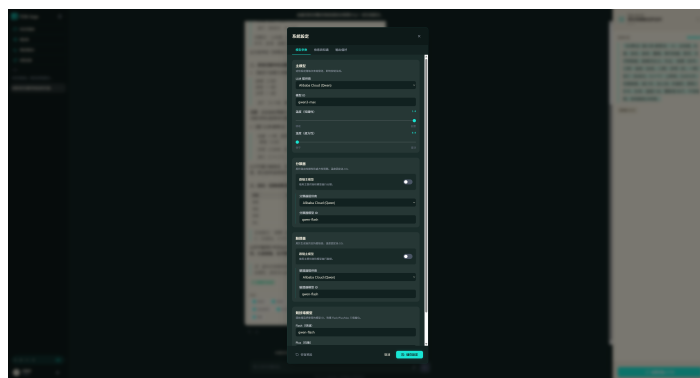


Figure 4.6: The Settings panel allowing configuration of LLM provider, model, and temperature.

4.8 Implementation Challenges and Solutions

4.8.1 Retrieval Ranking Bias

Challenge: Simple canonical formula definitions found in the Shanghan Lun were ranked lower than lengthy discourses found in later works from the Ming Dynasty such as Qianjin Fang. **Solution:** Primary sources have been prioritized over secondary commentaries, of length be it short. We have implemented Formula-Aware Canonical Retrieval and Source Authority Boost.

4.8.2 RAG Context Engineering

Challenge: Models like Qwen-turbo tended to ignore formatting rules when the retrieved context exceeded 2,000 characters. **Solution:** We switched to “Post-context anchoring,” moving the formatting instructions to the end of the prompt, and used Pattern Priming by styling the context as Markdown.

4.8.3 Diagnostic Over-Assertion

Challenge: Early versions would give definitive diagnoses (e.g., “You have Spleen Deficiency”), which is clinically irresponsible. **Solution:** We added Hedging Principles into the system prompt so the LLM uses uncertain language (e.g., “The symptoms suggest a possibility of...”) when evidence is limited, plus a mandatory medical disclaimer on all outputs.

Chapter 5

Evaluation

In this chapter we assess TCM-Sage’s performance relative to that of a non-specialized large language model (LLM) baseline. We present results from blind A/B testing using the “Arena” framework, with feedback from TCM practitioners and students.

5.1 Experimental Setup

The evaluation framework was designed to isolate the impact of the specialized RAG (Retrieval-Augmented Generation) pipeline and the TCM-specific system prompt.

5.1.1 System Configurations

Two distinct system configurations were compared:

1. **Treatment (TCM-Sage):** The full pipeline described in Chapter 4, including specialized RAG context from 17 classical texts, SymMap v2.0 knowledge graph integration, and a professional TCM system prompt.
2. **Baseline (Plain LLM):** A generic configuration using the same underlying Qwen model but with a “helpful assistant” prompt. To simulate a modern AI’s capabilities, the baseline had access to **DuckDuckGo Web Search** (5 results), representing the current standard for general-purpose retrieval.

5.1.2 Blind A/B Design

The evaluation was carried out blind and 59 TCM students and practitioners cast their votes.

- **Randomization:** For every query the positions of the Treatment and Baseline responses (A/B) were randomly mixed up.
- **Interaction:** Users entered a clinical or theoretical question and viewed corresponding results in a side by side format.
- **Metrics:** After reading both responses, voters chose from the options “A is Better,” “B is Better,” or “Tie”, which were then recorded in JSONL format for statistical analysis.

5.2 Quantitative Results

The primary goal of the quantitative analysis was to determine if the TCM-Sage system provided a statistically significant improvement over a general-purpose LLM.

5.2.1 Comparative Performance

Here is a breakdown of how the 59 votes were distributed.

- **TCM-Sage Wins:** 38 (64.4%)
- **Plain LLM Wins:** 15 (25.4%)

- **Ties:** 6 (10.2%)

We find that in over 60% of cases, TCM-Sage outperforms the baseline, more than doubling the win rate of the general-purpose search-augmented model.

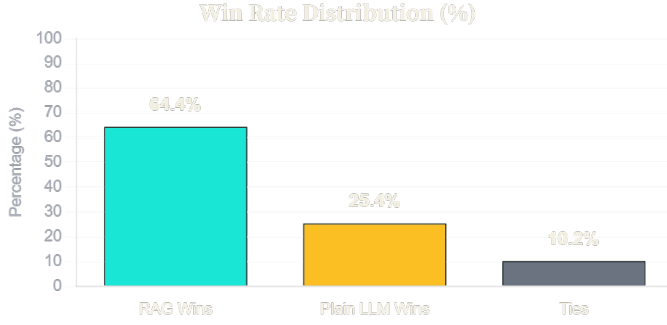


Figure 5.1: Win rate distribution across the three outcomes.

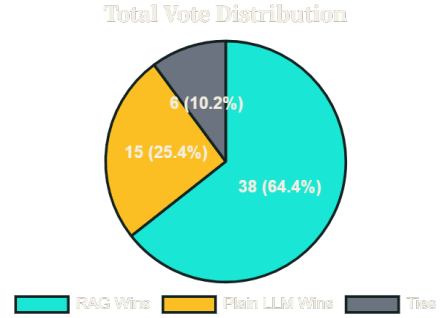


Figure 5.2: Total vote distribution (N=59).

5.2.2 Statistical Significance

To test the null hypothesis that there is no difference between TCM-Sage and the Plain LLM, a **paired t-test** was performed on the individual votes. Each vote was scored 1 for TCM-Sage and 0 for Plain LLM; ties were scored 0.5 for both systems.

- **t-statistic:** 3.44
- **p-value:** 0.0011

With $p < 0.01$, the results are highly significant, allowing us to reject the null hypothesis.

5.2.3 Effect Size

Cohen's d was calculated to measure the magnitude of the improvement:

- **Cohen's d:** 0.45

An effect size of 0.45 indicates a *medium effect*, suggesting that while the general-purpose model is a capable baseline, the specialized TCM-Sage system provides a meaningful and consistent advantage in the quality of clinical reasoning and source accuracy.

5.3 Temporal Analysis

The evaluation was conducted over a four-day period (April 3 to April 6, 2026), during which the system underwent minor iterative refinements based on initial feedback.

- **April 3:** 2 RAG / 1 Plain / 0 Ties (3 votes) — Initial testing phase.
- **April 4:** 13 RAG / 8 Plain / 6 Ties (27 votes) — Main testing batch from student groups.
- **April 5:** 8 RAG / 0 Plain / 0 Ties (8 votes) — Post system prompt and retrieval improvements.
- **April 6:** 12 RAG / 6 Plain / 0 Ties (18 votes) — Extended testing with a wider group of practitioners.
- **April 7:** 3 RAG / 0 Plain / 0 Ties (3 votes) — Continued testing.

The cumulative trend shows a consistent preference for TCM-Sage, with a notable peak in performance after specific clinical knowledge (e.g., unit conversions) was hard-coded into the system prompt.

5.4 Qualitative Feedback and Case Studies

Qualitative feedback from TCM professionals provided deeper insights into the specific strengths and weaknesses of both systems.

5.4.1 Dosage and Unit Conversion

A doctoral student at Guangzhou University of Chinese Medicine, who also practices at the Guangdong Provincial Hospital of TCM, tested a query regarding the dosage of *Mahuang Tang*. He observed that the general LLM (Baseline), aided by its web search capability, correctly identified the modern unit conversion for Han Dynasty weights. By contrast, the TCM-Sage system at that time provided an incorrect conversion (1 *liang* $\approx$ 3g instead of $\approx$ 13.8g), because the classical texts in the knowledge base do not contain modern unit conversion data.

Outcome: This feedback led to the immediate implementation of the *Measurement Conversion* table (1 *liang* $\approx$ 13.8g) in the system prompt. Subsequent tests showed that practitioners found the corrected system far superior to the baseline, as it provided both historical accuracy and modern relevance.

5.4.2 Source Traceability

Students consistently highlighted the **Citation Panel** as the most valuable feature. One student noted: “The ability to see the original classical text immediately, without having to copy-paste the formula name into a search engine to verify the source, is the key value of this system.” This indicates that the *traceability* provided by RAG is as important as the *content* of the generated response.

5.4.3 Diagnostic Hedging and Safety

A doctoral student who also practices as a physician noted that the early version of the system was “over-assertive” in its diagnosis of *Pi Xu* (Spleen Deficiency) based on limited symptoms. Following this, **Hedging Principles** were added to the prompt, requiring the system to use more tentative language (“The symptoms suggest a possibility of...”) when data is incomplete. The practitioner later confirmed that the updated responses felt more like a professional consultation than a “black-box” diagnosis.

5.4.4 Novel Clinical Insights

In one recorded case, a student consulted the system for a sore throat. The system suggested *Jiegeng Tang*, a classical formula the student had not considered. Upon reviewing the provided citations, the student confirmed the validity of the recommendation. This demonstrates the system’s potential as a “clinical co-pilot” that can surface relevant but overlooked classical knowledge.

5.5 Limitations of Evaluation

Although confirmed to be statistically significant, the findings have some limitations.

1. **Sample Size:** 59 votes are sufficient for statistical significance but do not represent a comprehensive clinical trial.
2. **Voter Demographic:** The majority of the voters were TCM students rather than senior clinicians with many decades of clinical experience.
3. **Evaluation Format:** The Arena tests overall user preference, which includes factors like formatting and tone, not just clinical accuracy.
4. **Prompt Influence:** Because the system prompts differed between the two models, the evaluation measures the effectiveness of the *entire pipeline* (Prompt + RAG + KG), not just the retrieval component in isolation.

5.6 Discussion of Results

Our evaluation results show that a specialized TCM RAG system significantly outperforms a general-purpose search-augmented LLM.

- **Accuracy vs. Verifiability:** The primary advantage of TCM-Sage lies in its ability to provide verbatim classical citations. While the general LLM can often “summarize” TCM knowledge from the web, it frequently hallucinates formula origins or misinterprets classical syntax.
- **The Value of Domain Knowledge:** The integration of a specialized knowledge graph and a professional system prompt (including ancient-to-modern measurement conversions) is essential for clinical utility. Without these, even a high-quality RAG system is prone to fundamental errors in interpretation.
- **Impact on the Field:** The $p = 0.0011$ result strongly suggests that specialized AI tools can provide a higher standard of information for the TCM community than general-purpose assistants, particularly in the areas of citation traceability and classical methodology adherence.

Chapter 6

Discussion and Contributions

6.1 Comparison with Existing TCM AI Systems

While all other TCM AI products pursue an “AI Doctor” paradigm, TCM-Sage functions as an evidence-synthesis reference tool. This distinction is not merely semantic; it reflects fundamentally different design philosophies with practical consequences.

General-purpose LLMs such as ChatGPT and Qwen operate as black boxes. When queried about TCM topics, they generate responses from parametric memory without any mechanism for source verification. A practitioner receiving a formula recommendation from ChatGPT cannot determine whether that recommendation originates from *Shanghan Lun*, from a modern textbook, or from the model’s interpolation of training data. In clinical contexts where incorrect dosage or formula selection carries real risk, this unverifiability is the primary obstacle to adoption, not hallucination per se.

Specialized TCM AI products have attempted to address domain accuracy but not transparency. HuatuoGPT achieved state-of-the-art results on multiple medical benchmarks [Zhang et al. \(2023\)](#). Qihuang 2.0 (Shuzhi Qihuang), a 32-billion-parameter multimodal model [East China Normal University School of Pharmacy \(2024\)](#), scored 88.1% on the TCM licensing examination and incorporated tongue and pulse image analysis. iFlytek’s TCM assistant was deployed across primary care institutions in Anhui Province [China National Radio \(CNR\) \(2024\)](#). These systems share a common “AI Doctor” design philosophy, focusing on direct diagnosis and prescription. TCM-Sage takes a fundamentally different direction: evidence synthesis with traceable citations from classical texts.

TCM-Sage addresses these barriers through five design decisions:

1. **Citation Traceability:** Every claim links to a specific classical text passage, visible through the Citation Panel. Practitioners can verify sources without copy-pasting text into external search tools.
2. **Classical Text Specialization:** The 17-text corpus provides depth in foundational TCM theory that general LLMs cannot match, backed by clause-level retrieval for structured texts.
3. **Full Local Deployment:** The entire pipeline, including embedding models and LLM inference, can run on consumer hardware via Ollama or LMStudio. No patient data leaves the institution’s network.
4. **Modular Knowledge Base:** Adding or removing knowledge sources requires only text ingestion, not model retraining. When practitioners suggested adding modern clinical texts during testing, the response was immediate: the architecture supports this without modification.
5. **Ancient Measurement Accuracy:** An embedded conversion table provides historically verified unit conversions (Han dynasty *liang* $\approx$ 13.8g, not the commonly misquoted 3g), preventing dosage calculation errors that other systems produce.

The SymMap v2.0 knowledge graph deserves specific mention. Unlike the classical texts, SymMap is a modern pharmacological database integrating contemporary research. The crosswalk bridge connects these modern entities to classical terminology, creating a bidirectional link between historical medical knowledge and current pharmacological understanding. This hybrid of classical texts and modern knowledge graph is unique among TCM AI systems.

6.2 Strengths Demonstrated

The Arena evaluation provides quantitative evidence: with $p = 0.0011$ and Cohen’s $d = 0.45$, the preference for TCM-Sage over the baseline is statistically significant at the 1% level with a medium effect size. This result was observed across five days of testing with different tester groups.

Qualitative feedback reinforced specific strengths. A TCM student consultant summarized four key advantages observed during extended testing:

1. **Focused, relevant responses:** The system provides adequately elaborated answers with valid arguments while avoiding irrelevant details. Unlike general LLMs that tend to produce verbose, unfocused responses mixing TCM with Western medicine or unrelated topics, TCM-Sage stays within the classical framework.
2. **Credible source extraction and integration:** The system extracts and integrates credible sources directly from TCM classics, presenting them alongside the generated response through the Citation Panel.
3. **TCM-perspective clinical guidance:** The system guides practitioners in thinking from TCM perspectives, including *Materia Medica* recommendations and treatment approach demonstrations grounded in classical methodology (*Bian Zheng Lun Zhi*).
4. **Practitioner-informed tuning:** The system prompt and retrieval pipeline were iteratively refined based on feedback from TCM students, resulting in output that aligns with how practitioners expect clinical information to be structured.

The Citation Panel received the strongest individual endorsement, with testers describing it as eliminating the need to manually verify AI-generated claims against source texts. In one case, the system recommended *Jiegeng Tang* (Platycodon Decoction) for a sore throat that the consulting student had not considered. Upon review, he confirmed this as a valid and potentially superior approach from a classical perspective, demonstrating the system’s ability to surface overlooked classical formulas.

The modular architecture received indirect validation when practitioners at Guangdong Provincial Hospital of TCM suggested expanding the corpus to include modern clinical masters’ works (Zhang Xichun’s *Yi Xue Zhong Zhong Can Xi Lu*, Shi Xuemin’s acupuncture texts, and Dong’s Extraordinary Points system). The fact that such expansion is architecturally straightforward confirms that the design decision to use RAG over fine-tuning was sound.

6.3 Limitations and Honest Assessment

Several limitations emerged during testing and must be acknowledged.

Knowledge Boundary Errors. The system’s accuracy is bounded by its corpus. When a practitioner asked about *Mahuang Tang* dosage, the system stated that one Han dynasty *liang* equals approximately 3 grams, a significant error (the correct value is 13.8–15.6g based on archaeological evidence). This occurred because the classical texts contain dosage ratios but not modern unit conversions. The plain LLM baseline, with access to web search, retrieved the correct conversion. This limitation was addressed by embedding a historically verified measurement conversion table directly into the system prompt.

Corpus Coverage Gaps. The 17-text corpus covers foundational theory, *Shanghan/Wenbing*, herbology, formulary, and acupuncture. It does not cover surgery, orthopedics, ophthalmology, specialized gynecology or pediatrics texts, or pulse diagnosis monographs. Practitioner feedback was direct: “These texts are too traditional. Real clinical learning requires modern clinical masters’ experience and knowledge systems.” This feedback, while critical, validates the system’s modular design as the appropriate architecture for iterative corpus expansion.

Diagnostic Over-Assertion. The system initially presented diagnostic interpretations with excessive confidence. When analyzing a pediatric case, it concluded *Pi Xu* (spleen deficiency) without sufficient evidence in the case data, and classified an initial rash presentation as *Re Yu Yu Ying* (heat trapped in the nutritive level) when the clinical presentation better supported *Xie Qi Zai Biao* (pathogen at the exterior). Hedging principles were added to the system prompt to mitigate this tendency, but the underlying challenge of calibrating LLM confidence in medical reasoning remains partially unresolved.

Evaluation Scale. The 59-vote Arena sample, while statistically sufficient for significance testing, represents a limited evaluator pool. Most voters were TCM students rather than senior practitioners. The Arena

format measures overall preference but does not decompose quality into specific dimensions such as citation accuracy, diagnostic reasoning, or readability.

6.4 Ethical Considerations

TCM-Sage is positioned as an evidence reference tool, not a diagnostic system. This distinction carries ethical weight. The system does not claim to diagnose, and its outputs carry an explicit disclaimer that all suggestions are for clinical reference only, with final diagnostic and prescriptive authority belonging to licensed practitioners.

Data privacy is addressed architecturally. Full local deployment eliminates the need to transmit patient data to external servers. During testing, practitioners at Guangdong Provincial Hospital of TCM noted this as a meaningful advantage, given institutional policies restricting patient data sharing with cloud AI services.

The use of AI tools in developing this project is disclosed in the Acknowledgments section, in compliance with university policy.

Table 6.1: Feature comparison between TCM-Sage and existing TCM AI systems.

Feature	TCM-Sage	HuatuoGPT	Qihuang 2.0	iFlytek TCM
Primary Approach	Evidence synthesis	AI Doctor (diagnosis)	AI Doctor (multi-modal)	AI Doctor (primary care)
Citation Traceability	Clause-level, click-through	None reported	None reported	None reported
Knowledge Base	17 classical texts + SymMap KG	Fine-tuned on medical corpus	TCM KG + tongue/pulse images	Expert KG + clinical data
Local Deployment	Full local (LLM + embedding)	Cloud only	Cloud only	Cloud only
Knowledge Update	Add text files, re-ingest	Retrain model	Retrain model	Retrain model
Evaluation	Arena blind A/B ($p = 0.0011$)	Medical benchmark SOTA	TCM licensing exam (88.1%)	9,800+ assisted diagnoses

Chapter 7

Conclusion and Future Work

7.1 Summary of Contributions

In this project we developed and evaluated a working evidence-synthesis system for classical Traditional Chinese Medicine (TCM) texts, called TCM-Sage. Classical TCM texts present special challenges to Language Modeling and LLMs in general. Currently there are few effective and robust AI products addressing TCM-related needs, and none of them use classical medical texts. We addressed a particular gap in this domain by developing reliable classical medical text retrieval with full citations and source tracking as the core capability of TCM-Sage.

The principal technical contributions are:

1. **Hybrid RAG Pipeline for Classical Chinese:** We present a hybrid RAG pipeline that integrates DashScope text-embedding-v4 (1024 dimensions) vector search with SymMap v2.0 knowledge graph traversal. To bridge the large semantic drift over two millennia of medical writings, we build a crosswalk between modern medical terms and their classical Chinese counterparts.
2. **Domain-Specific Retrieval Optimizations:** Formula-aware canonical retrieval ensures that queries about classical formulas (e.g., *Mahuang Tang*) return results from their canonical source texts first. Source authority boosting applies topic-dependent chronological weighting. Clause-level chunking preserves the natural semantic units of structured texts such as *Shanghan Lun* (388 clauses) and *Jingui Yaolue* (489 clauses).
3. **RAG Context Engineering:** The discovery of Pattern Priming — that formatting retrieved context as markdown causes the LLM to mirror markdown in its output — and post-context instruction anchoring, where formatting directives placed after retrieved passages maintain effectiveness despite the “lost-in-the-middle” attention pattern.
4. **Arena Blind A/B Evaluation:** A dual-streaming evaluation system inspired by the LMSYS Chatbot Arena, with paired t-test and Cohen’s d validation. Results from 59 votes yielded $p = 0.0011$ and $d = 0.45$, establishing statistically significant preference for the RAG system over a general LLM with web search.
5. **Practitioner Validation:** Testing by TCM students and doctoral candidates at Guangzhou University of Chinese Medicine (several of whom also practice at Guangdong Provincial Hospital of TCM) confirmed the system’s clinical utility for evidence retrieval. The Citation Panel was identified as the primary value differentiator, and the system demonstrated the ability to surface classical formulas that practitioners had not considered.

7.2 Future Work

Practitioner feedback during testing defined a clear development roadmap. The following directions are ordered by priority and feasibility.

7.2.1 Corpus Expansion

The most consistent feedback was that the current 17-text corpus, while strong in classical theory, lacks modern clinical practice material. Specific texts recommended by practitioners include:

- **Modern clinical masters:** Zhang Xichun’s *Yi Xue Zhong Zhong Can Xi Lu* (a foundational text for Chinese-Western integrated medicine), works by Liu Duzhou and Feng Shilun (classical formula specialists), and Zhang Jingyue’s *Jingyue Quanshu*.
- **Modern acupuncture:** Shi Xuemin’s *Shi Yong Zhen Jiu Xue* (evidence-based stroke treatment protocols), Dong’s Extraordinary Points system, Cheng Dan’an’s *Zhongguo Zhen Jiu Xue* (integrating Western neuroanatomy), and Takeda Ken’s *Zhen Jiu Zhen Sui* (Japanese acupuncture).
- **Pharmacology:** *Ben Cao Yao Zheng* for clinical drug evidence.
- **Chinese-Western integration:** Modern clinical pharmacology references to support practitioners who combine both approaches.

The modular RAG architecture makes corpus expansion straightforward: new texts require only UTF-8 source files and a single ingestion run. No model retraining is necessary.

7.2.2 Platform Development

A TCM student tester expressed interest in recommending the system to his university’s School of Chinese Medicine and exploring case competition entry. This suggests potential for development beyond an individual project into a platform serving TCM education programs. Each institution could maintain its own knowledge base supplemented with course materials, clinical SOPs, and local reference documents.

7.2.3 Patient-Friendly Mode

Practitioners noted that pure TCM terminology is difficult for general audiences to understand. A future mode could provide simplified explanations alongside classical citations, expanding the user base to include patients and health-conscious individuals seeking reliable TCM information.

7.2.4 Fine-Tuned Base Model Integration

The current architecture uses general-purpose Qwen models as the generation backbone. Stacking a TCM-specialized fine-tuned model (such as HuatuoGPT or a custom fine-tune) as the base, with the RAG pipeline providing retrieved context, could yield compounding quality improvements. The RAG architecture explicitly supports this: the base LLM is a configurable component.

7.2.5 Expanded Evaluation

Future evaluation should address the current study’s limitations: larger sample sizes, inclusion of senior practitioners as evaluators, and dimension-specific scoring rubrics that separately measure citation accuracy, diagnostic reasoning quality, completeness, and readability. Longitudinal studies tracking system improvement across development iterations would strengthen the evidence base.

7.3 Goal Completion

The original project proposal set out to build “an explainable, evidence-backed tool” based on RAG architecture that would function as “an intelligent clinical reference assistant” with traceable citations, hybrid retrieval (vector search and knowledge graph), and user acceptance testing with TCM practitioners. The following table summarizes the completion status of each proposed aim:

Table 7.1: Proposed aims and completion status.

Proposed Aim	Status	Notes
Evidence-synthesis tool, not AI Doctor	Achieved	Core design principle maintained
Glass Box citations linked to source text	Achieved	Citation Panel with click-through
RAG architecture with hybrid retrieval	Achieved	Vector + KG + reranker
Specialized TCM knowledge base	Exceeded	17 texts (proposal suggested 1)
Knowledge graph for terminology bridging	Achieved	SymMap v2.0 (18,450 entities)
User acceptance testing with practitioners	Achieved	Arena blind A/B (59 votes, $p = 0.0011$)

The evaluation methodology evolved from the initially proposed TCMEval-SDT benchmark and Likert-scale surveys to a blind A/B Arena system with statistical validation, following discussions with the project supervisor. This approach better captures overall system quality as perceived by end users, rather than isolated retrieval metrics.

All core proposed aims have been achieved or exceeded. The system is functional, tested with real practitioners, and statistically validated.

7.4 Closing Remarks

TCM-Sage demonstrates that retrieval-augmented generation is a viable approach for making classical medical literature accessible through modern AI interfaces. The system does not replace human expertise; it provides the evidence base that human experts need to make informed decisions. In a field where the source texts span two thousand years of accumulated clinical wisdom, the ability to search, retrieve, and cite that wisdom accurately and transparently has practical value.

The project began with a straightforward observation: practitioners need better tools for navigating classical texts. It concludes with a working system, statistical evidence of its effectiveness, and a clear roadmap shaped by the practitioners it was built to serve.

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